Lab Task 08

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Subject	**SCD**

Part A

```java
public class Employee {
public static void main(String[] args) {
String employeeName1 = "Ali";
double baseSalary1 = 50000;
double bonus1 = baseSalary1 * 0.5;
double taxDeduction1 = baseSalary1 * 0.25;
System.out.println("Salary Details for " + employeeName1 + ":");
System.out.println("Base Salary: " + baseSalary1);
System.out.println("Bonus: " + bonus1);
System.out.println("Tax Deduction: " + taxDeduction1);
String employeeName2 = "Sara";
double baseSalary2 = 60000;
double bonus2 = baseSalary2 * 0.3;
double taxDeduction2 = baseSalary2 * 0.2;
System.out.println("Salary Details for " + employeeName2 + ":");
System.out.println("Base Salary: " + baseSalary2);
System.out.println("Bonus: " + bonus2);
System.out.println("Tax Deduction: " + taxDeduction2);
}
}
```

Smells Identified:

Code Duplication: The same calculation and printing logic is repeated for each employee.

Poor Organization: All the logic is in the main method with no separation of concerns.

Hardcoded Values: Bonus and tax rates are hardcoded directly in the calculations. If the bonus rate changes from 0.5 to 0.6, you must manually update every occurrence in the code.

Lack of Abstraction: No proper class structure to represent an Employee and its behaviors.

Refactored Code

```
public class Employee {
   private final String name;
   private final double baseSalary;
   private final double bonusRate;
   private final double taxRate;


   public Employee(String name, double baseSalary, double bonusRate, double taxRate) {
      this.name = name;
      this.baseSalary = baseSalary;
      this.bonusRate = bonusRate;
      this.taxRate = taxRate;
   }


   public double calculateBonus() {
      return baseSalary * bonusRate;
   }


   public double calculateTaxDeduction() {
      return baseSalary * taxRate;
   }


   public void printSalaryDetails() {
      System.out.println("Salary Details for " + name + ":");
      System.out.println("Base Salary: " + baseSalary);
      System.out.println("Bonus: " + calculateBonus());
      System.out.println("Tax Deduction: " + calculateTaxDeduction());
   }


   public static void main(String[] args) {
      Employee ali = new Employee("Ali", 50000, 0.5, 0.25);
      Employee sara = new Employee("Sara", 60000, 0.3, 0.2);

      ali.printSalaryDetails();
```

```
        System.out.println();
        sara.printSalaryDetails();
    }}
```

Part B

```
public class InvoicePrinter {
public void printInvoice(String customerName, int[] items) {
System.out.println(&quot;Customer: &quot; + customerName);
int total = 0;
for (int item : items) {
System.out.println(&quot;Item: &quot; + item);
total += item;
}
double tax = total * 0.1;
System.out.println(&quot;Tax: &quot; + tax);
System.out.println(&quot;Total with Tax: &quot; + (total + tax));
}
public static void main(String[] args) {
int[] items = {100, 200, 300};
new InvoicePrinter().printInvoice(&quot;Alice&quot;, items);
}
}
```

Refactored Code

```
public class InvoicePrinter {
    public void printInvoice (String customerName, int[] items) {
        printCustomerHeader (customerName);
        int subtotal = calculateAndPrintItems(items);
        printTaxAndTotal(subtotal);
    }


    private void  printCustomerHeader(String customerName) {
        System.out.println ("Customer: " + customerName);
    }


    private int  calculateAndPrintItems(int[] items) {
        System.out.println("Items:");
        int total = 0;
```

```
        for (int item : items) {
            System.out.println("  - " + item);
            total += item;
        }
        return total;
    }


    private void printTaxAndTotal(int subtotal) {
        double tax = calculateTax(subtotal);
        double totalWithTax = subtotal + tax;

        System.out.println("Subtotal: " + subtotal);
        System.out.println("Tax (10%): " + tax);
        System.out.println("Total with Tax: " + totalWithTax);
    }


    Private double calculateTax(int amount) {
        return amount * 0.1;
    }


    public static void main(String[] args) {
        int[] items = {100, 200, 300};
        new InvoicePrinter().printInvoice("Alice", items);
    }
}
```